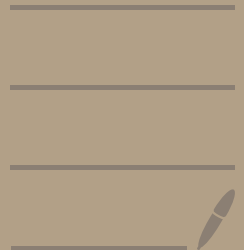


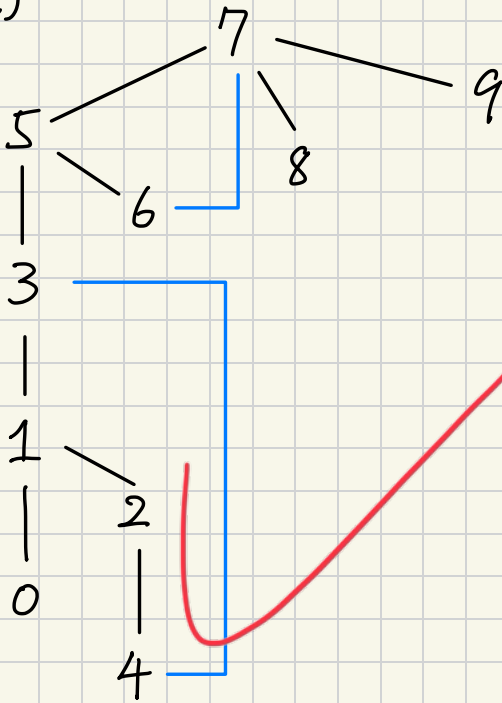
100

Homework 4

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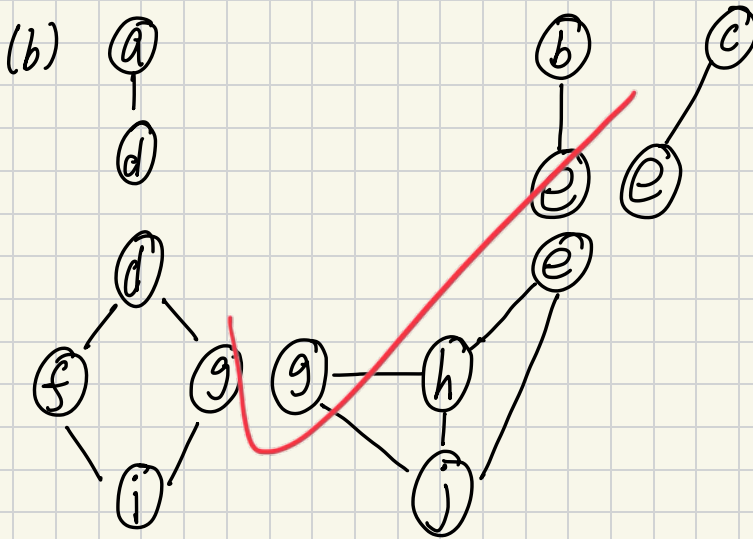
1. (a)



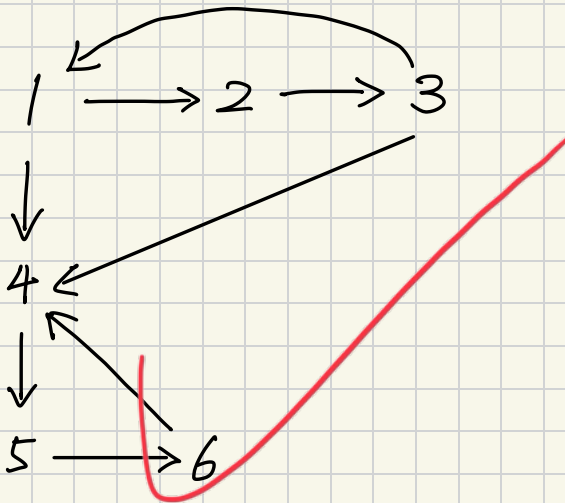
(b)

Vertex	0	1	2	3	4	5	6	7	8	9
dfn	4	3	5	2	6	1	7	0	8	9
low	4	2	2	2	2	0	0	0	8	9

2. (a) articulation points: d, e, g



3.



4. (a) dfs: 0 1 2 4 7 5 6 3

(b) bfs: 0 1 2 3 4 5 6 7

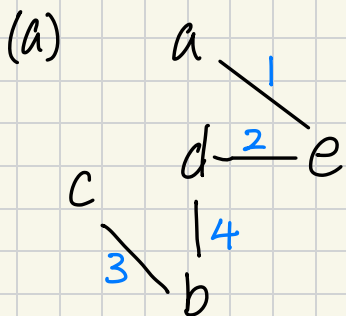
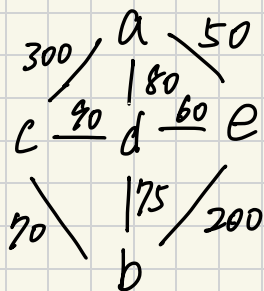
5. (a) bfs (v_1): v_1 v_2 v_7 v_3 v_4 v_5 v_6

(b) dfs (v_1): v_1 v_2 v_3 v_4 v_5 v_6 v_7

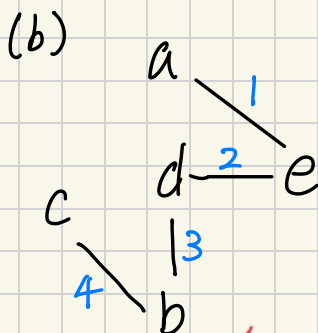
6

Iteration	Vertex Selected	Distance							
		A	B	C	D	E	F	G	H
initial	A	0	1	∞	∞	∞	∞	1	∞
1	B	0	1	5	∞	2	∞	1	∞
2	G	0	1	5	∞	2	∞	1	2
3	E	0	1	5	∞	2	3	1	2
4	H	0	1	5	∞	2	3	1	2
5	F	0	1	4	∞	2	3	1	2
6	C	0	1	4	5	2	3	1	2
7	D	0	1	4	5	2	3	1	2

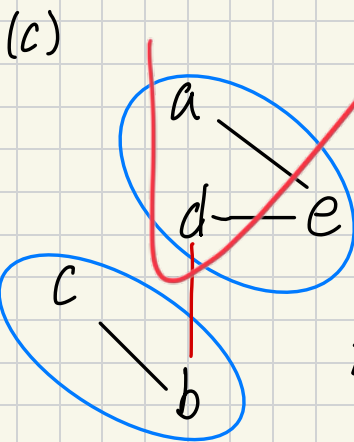
7.



total = 255



total = 255



total = 255

8.

K	$\text{dist}^k[u]$						
	0	1	2	3	4	5	6
1	0	6	5	5	∞	∞	∞
2	0	3	3	5	5	4	∞
3	0	1	3	5	2	4	7
4	0	1	3	5	0	4	5
5	0	1	3	5	0	4	3
6	0	1	3	5	0	4	3